Dimpled e/D comparison and operating trends

40 archived fits compared with 24 revised bounded fits — reading notes 1/2

Historical model: Archived direct-Colebrook thermal extension with no smooth-trained multiplier; historical fit bounds allowed e/D up to 1.

New model: Revised DH-relative-roughness extension plus smooth-trained baseline multiplier; fitted e/D restricted to 0..0.05.

Not a controlled single-change test: Comparison includes implementation changes AND empirical baseline correction.

Missing new fits: 13 targets below smooth baseline; 3 above e/D=.05 endpoint. Blank e/D is not zero. No fits outside tested bound are invented.

Selection bias: New e/D trends use only 24 matched cases. Missing cases depend on target and operating conditions; these are not representative complete-sample correlations.

Dimpled e/D comparison and operating trends

40 archived fits compared with 24 revised bounded fits — reading notes 2/2

Flux predictor: Heat flux on the trend x-axis is the nominal experimental 2000/3000 group, not exact measured flux. Numerical flux is a solver output.

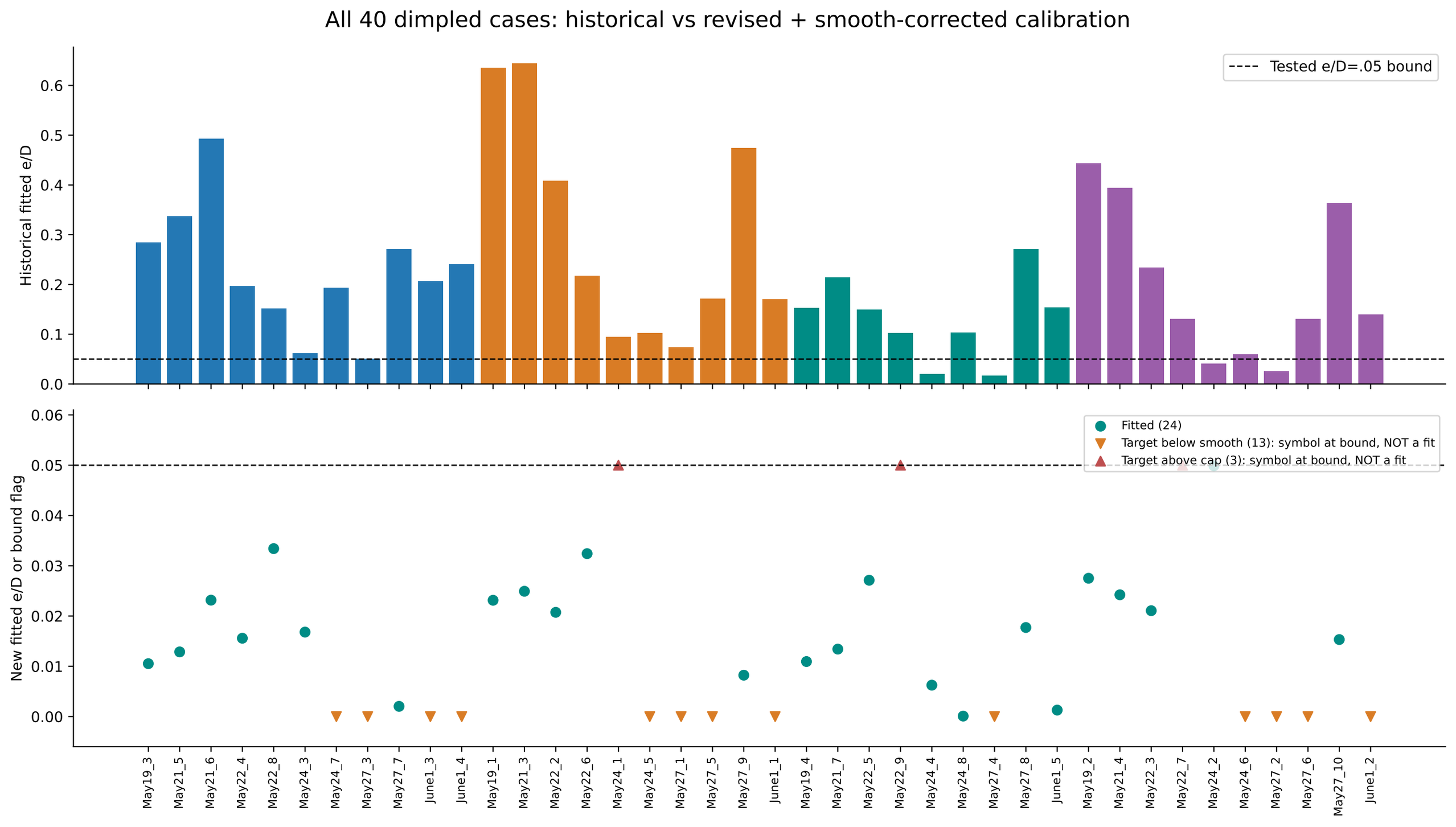
Mass flux: Plots use source-table reported mass flux; geometry-derived mass flux is also exported. m12/m15 are source group labels, not assumed exact mass fluxes.

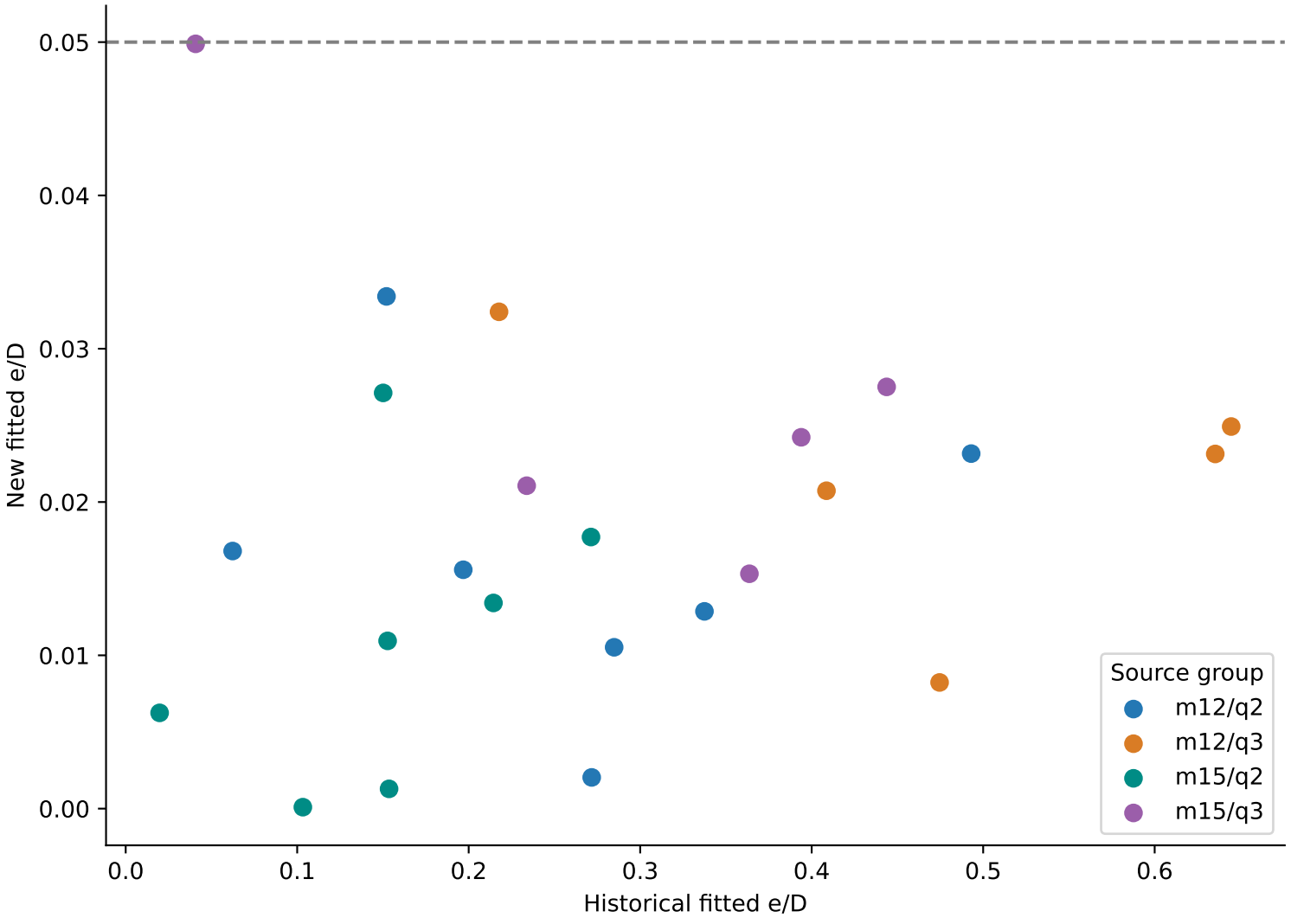
Built-in dependence: The smooth multiplier already uses temperature relative to Tpc, mass flux and nominal flux. New e/D trends partly reflect that correction and its residual errors.

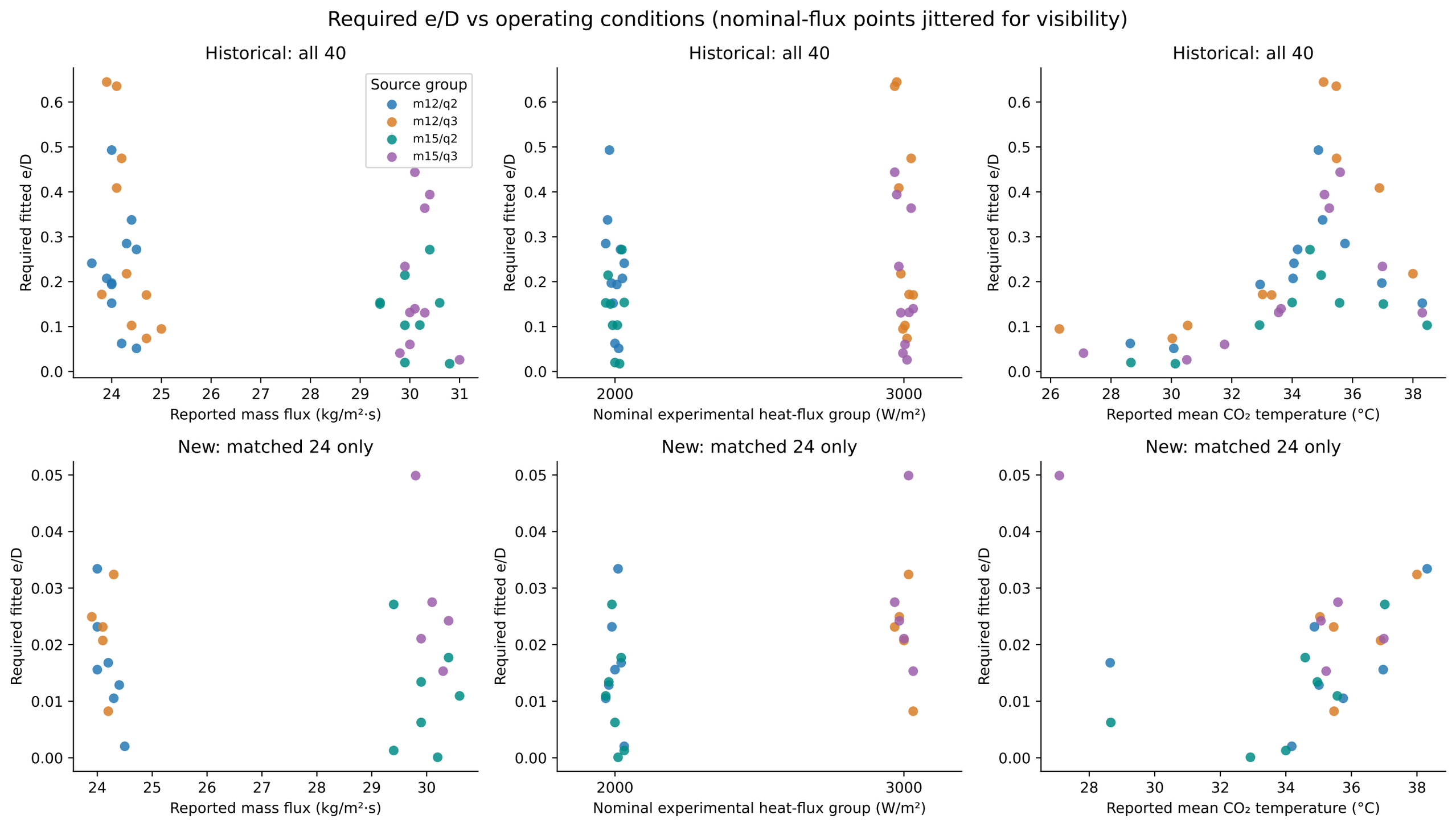
Interpretation: e/D values are fitted effective parameters. The 0.05 cap is a chosen tested bound, not proven physical roughness.

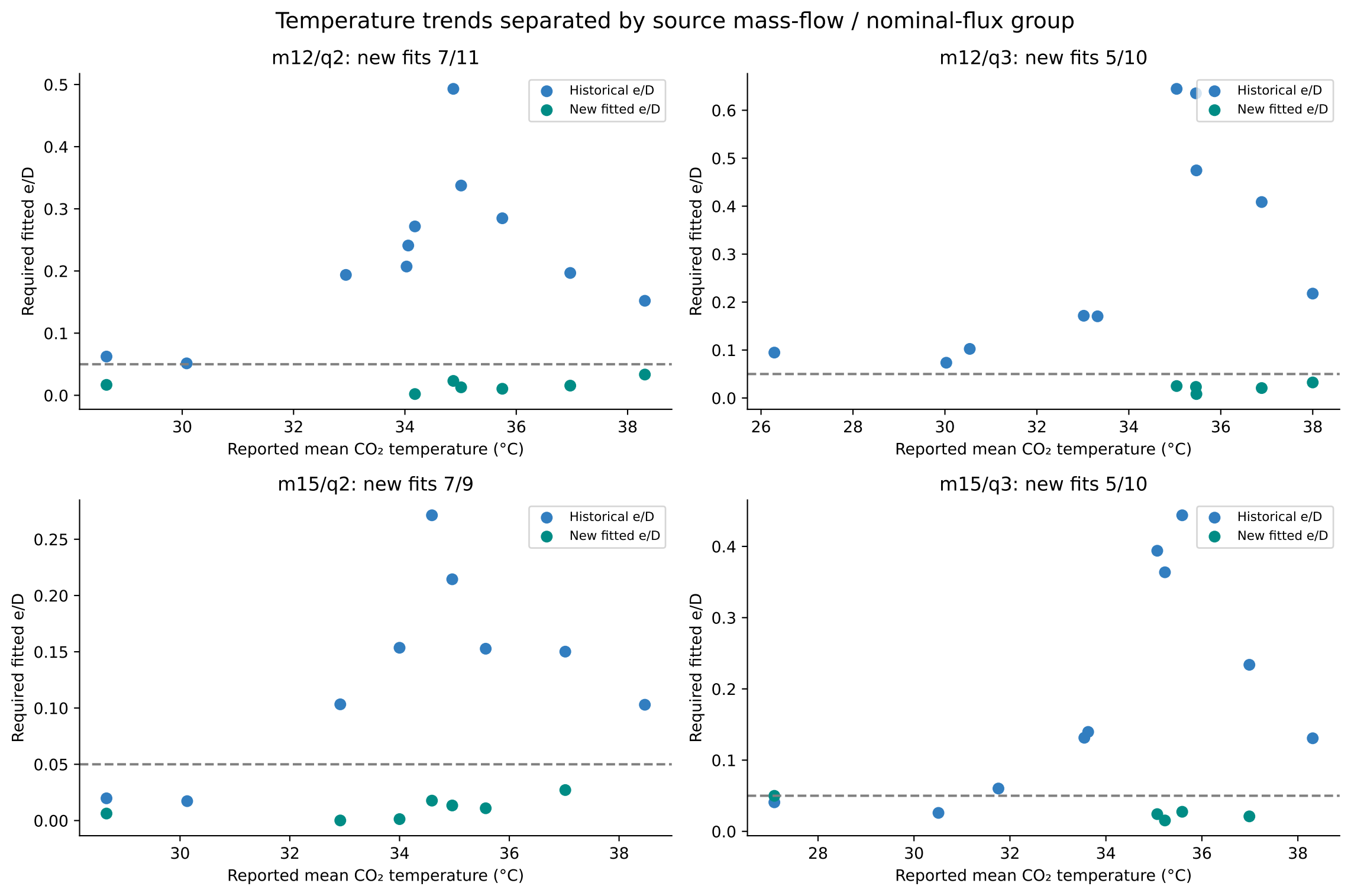
Statistics: Pearson and Spearman coefficients are exploratory associations, not validated laws, causal effects or independent predictive validation.

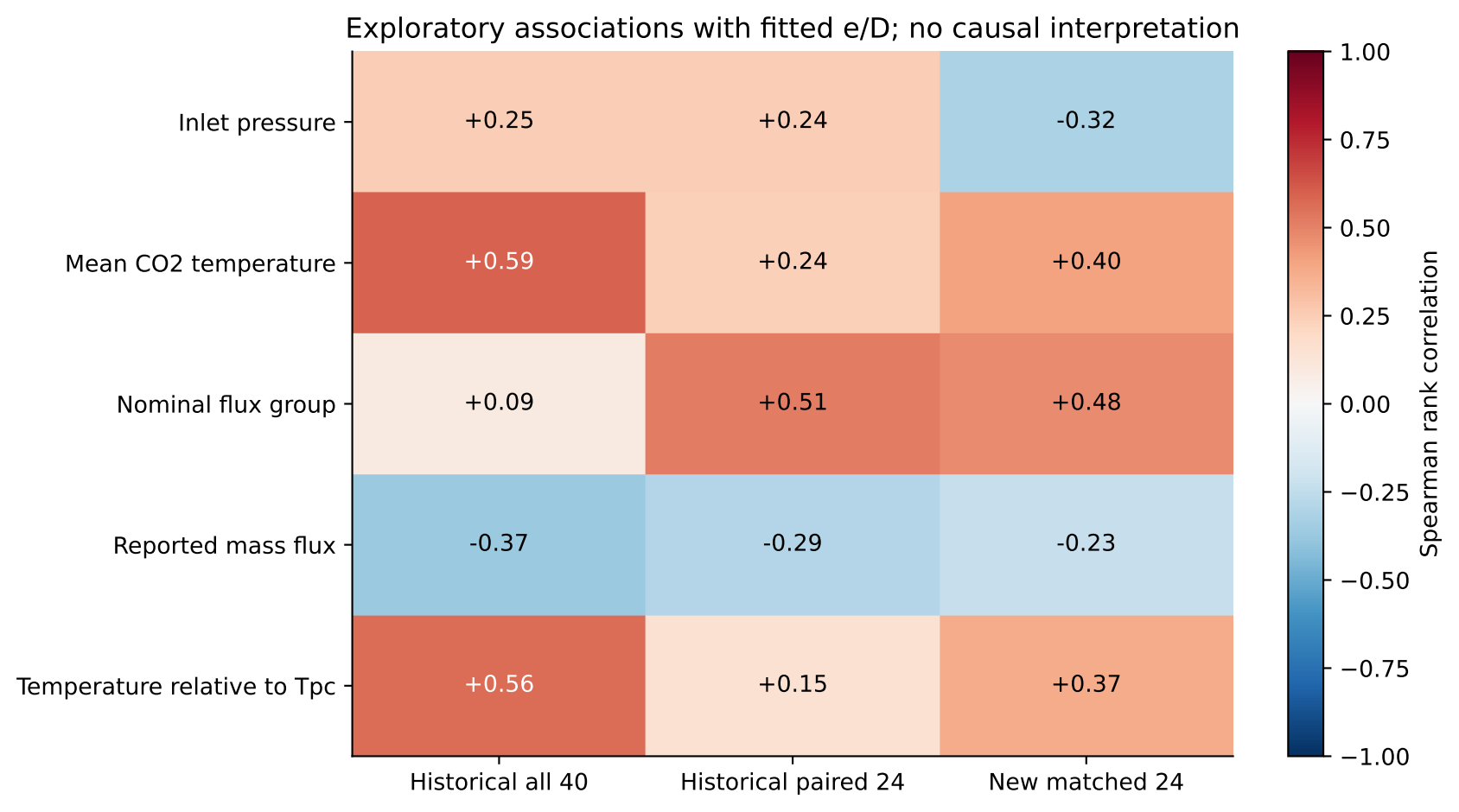
Sources: Model results snapshot da28165; results/Spencer-vs-Numerical-HTC-All-40-Cases.csv; results/smooth_baseline/Dimpled-transfer.summary.csv; Dimpled_cases.csv; experimental_targets.csv.

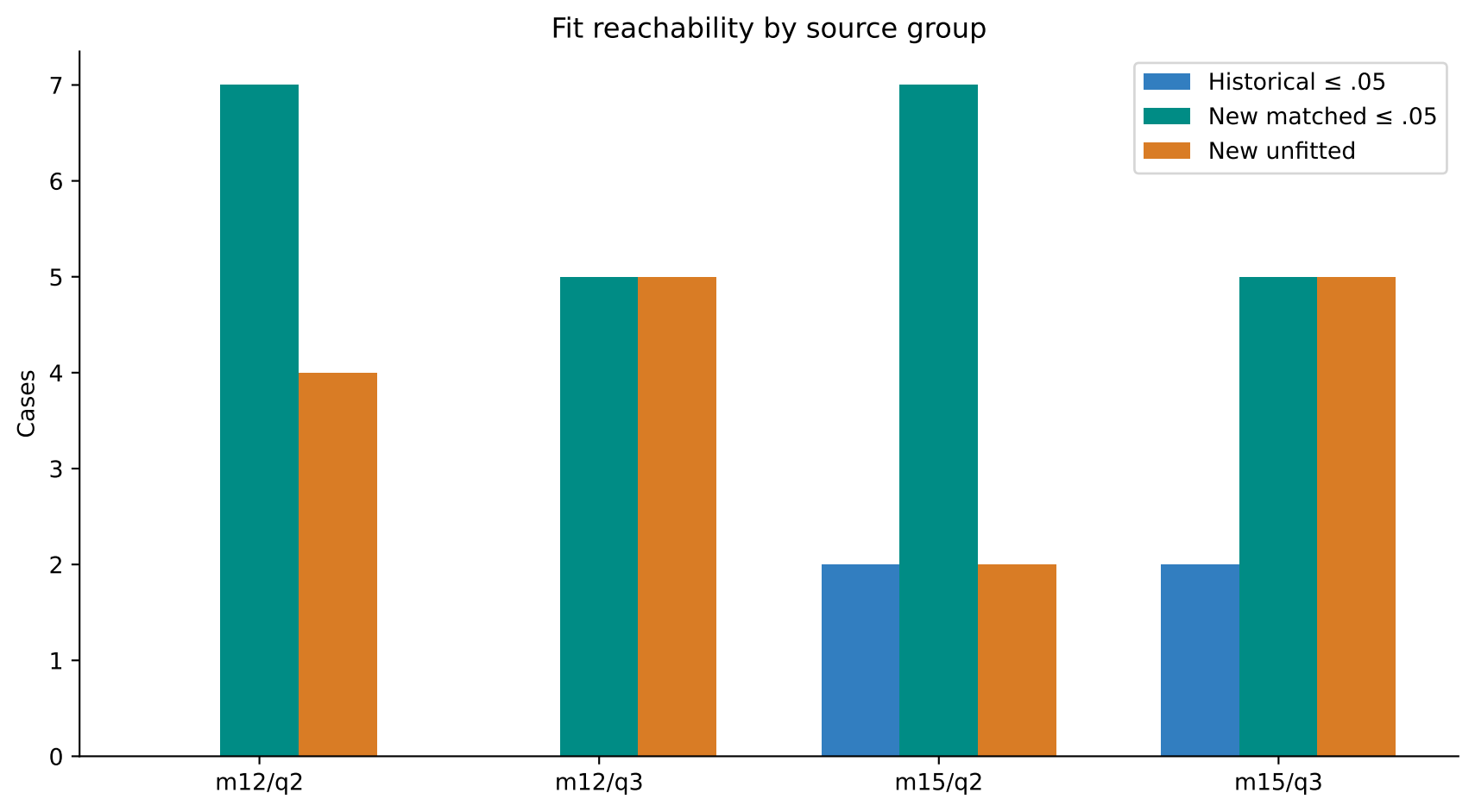

All 40 dimpled cases: historical vs revised + smooth-corrected calibration
Historical fitted e/D
Tested e/D=.05 bound
New fitted e/D or bound flag
Fitted (24)
Target below smooth (13): symbol at bound, NOT a fit
Target above cap (3): symbol at bound, NOT a fit
May19_3
May21_5
May21_6
May22_4
May22_8
May24_3
May24_7
May27_3
May27_7
June1_3
June1_4
May19_1
May21_3
May22_2
May22_6
May24_1
May24_5
May27_1
May27_5
May27_9
June1_1
May19_4
May21_7
May22_5
May22_9
May24_4
May24_8
May27_4
May27_8
June1_5
May19_2
May21_4
May22_3
May22_7
May24_2
May24_6
May27_2
May27_6
May27_10
June1_2


Paired comparison: 24 matched cases only
New fitted e/D
Historical fitted e/D
0.05
0.04
0.03
0.02
0.01
0.00
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Source group
m12/q2
m12/q3
m15/q2
m15/q3


Required e/D vs operating conditions (nominal-flux points jittered for visibility)
Historical: all 40
Historical: all 40
Historical: all 40
Source group
m12/q2
m12/q3
m15/q2
m15/q3
Required fitted e/D
Reported mass flux (kg/m²·s)
Nominal experimental heat-flux group (W/m²)
Reported mean CO₂ temperature (°C)
New: matched 24 only
New: matched 24 only
New: matched 24 only
Required fitted e/D
Reported mass flux (kg/m²·s)
Nominal experimental heat-flux group (W/m²)
Reported mean CO₂ temperature (°C)


Temperature trends separated by source mass-flow / nominal-flux group
m12/q2: new fits 7/11
m12/q3: new fits 5/10
m15/q2: new fits 7/9
m15/q3: new fits 5/10
Historical e/D
New fitted e/D
Required fitted e/D
Reported mean CO₂ temperature (°C)


Exploratory associations with fitted e/D; no causal interpretation
Inlet pressure
Mean CO2 temperature
Nominal flux group
Reported mass flux
Temperature relative to Tpc
+0.25
+0.24
-0.32
+0.59
+0.24
+0.40
+0.09
+0.51
+0.48
-0.37
-0.29
-0.23
+0.56
+0.15
+0.37
Historical all 40
Historical paired 24
New matched 24
Spearman rank correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Fit reachability by source group
Historical ≤ .05
New matched ≤ .05
New unfitted
Cases
0
1
2
3
4
5
6
7
m12/q2
m12/q3
m15/q2
m15/q3